Validation report

Interval quiz platform

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Accessibility run: 12 October 2026, 18:33:50 UTC

ABSTRACT

This report re-examines the two quiz workflows described in Labs 5–8. The retained Lab 9 browser runs completed the instructor task in 5/5 (100%) runs and the student task in 5/5 (100%) runs. The corresponding Lab 7 records show 0/7 (0%) and 0/7 (0%). A separate custom accessibility check recorded 49 passes and 6 failures across 6 screens. These findings support a narrower conclusion than a usability study: the scripted paths completed in the recorded environment. They do not establish how independently recruited people perform, or whether the interface conforms to WCAG.

01 Objective

Determine whether the revised instructor publishing flow and student submission flow can complete, then identify what remains unverified. Lab 7 provides the earlier execution record; Lab 8 documents the intervening changes. The evidence used here is the saved JSON output, the test harnesses, and the inspection notes [1–5].

02 Method and scope

Browser execution. Playwright drives the real interface. The identifiers N11, N12, N13, N14, N15 label fixture-backed runs, not documented human participants. Student accounts differ between runs; Task 1 reuses the configured instructor account. Isolated browser contexts separate session state.

Inspection. The existing notes compare ten Nielsen heuristics and record a walkthrough of the student path. Their ratings are qualitative judgements, reported separately from the measured browser outcomes.

Accessibility. A custom DOM and style checker samples six screens. No participant-session records or manual screen-reader results accompany these artifacts. The scope limits are detailed in Section 6.

03 Recorded browser outcomes

Task 1 creates a draft, adds four question types, saves rules, and publishes. Task 2 opens preflight, answers the four question types, and submits. Durations below are measured script execution times, including navigation, waits, and setup. Task 1 starts timing before fixture provisioning and sign-in—not when a person begins reading or making decisions.

Table 1. Retained Lab 9 runs. Errors and completed steps are copied from each task record.

Run ID	Task 1 Duration / errors / steps	Task 2 Duration / errors / steps
N11	12.2 s / 0 / 14/14 complete	6.0 s / 0 / 10/10 complete
N12	5.7 s / 0 / 14/14 complete	4.8 s / 0 / 10/10 complete
N13	6.5 s / 0 / 14/14 complete	4.7 s / 0 / 10/10 complete
N14	6.2 s / 0 / 14/14 complete	4.5 s / 0 / 10/10 complete
N15	6.1 s / 0 / 14/14 complete	5.0 s / 0 / 10/10 complete

Earlier and current records

Table 2. Aggregates computed from both saved results files by compare.py.

Measure	Lab 7	Lab 9
Task 1 completion	0/7 (0%)	5/5 (100%)
Task 2 completion	0/7 (0%)	5/5 (100%)
Task 1 median duration	38.2 s	6.2 s
Task 2 median duration	67.5 s	4.8 s
Task 1 mean recorded errors	2.0	0.0
Task 2 mean recorded errors	4.0	0.0

The observed completion-rate difference is +100 percentage points for Task 1 and +100 for Task 2. This is a descriptive comparison between retained runs, not a population estimate or a controlled causal measurement.

Timing caveat. Lab 7 records elapsed time for the whole failed run, including successive failed steps—not time to the first blocker. Lab 9 records completed-run elapsed time. These measures do not establish that people now finish faster.

04 Criteria and interpretation

The evaluation plan lists four numerical thresholds. Applying those inequalities to script output is useful as a regression check, but does not validate targets written for unassisted people. No claim of preregistration is made here.

Table 3. Arithmetic checks only. Human usability targets remain unassessed.

Criterion	Recorded observation	Numerical check
T1 · Task 1 completion ≥ 90%	5/5 (100%)	Within threshold
T2 · Task 2 completion = 100%	5/5 (100%)	Within threshold
T3 · Task 2 median ≤ 180 s; maximum ≤ 240 s	Median 4.8 s Maximum 6.0 s	Within threshold
T4 · Task 1 errors ≤ 1 in every run	Maximum 0 Total 0	Within threshold

The error limit is evaluated for each run, not only against the mean. A single run with more than one error must fail T4 even when other runs contain none. Fast automated execution cannot establish first-time usability or low cognitive effort.

Publishing controls

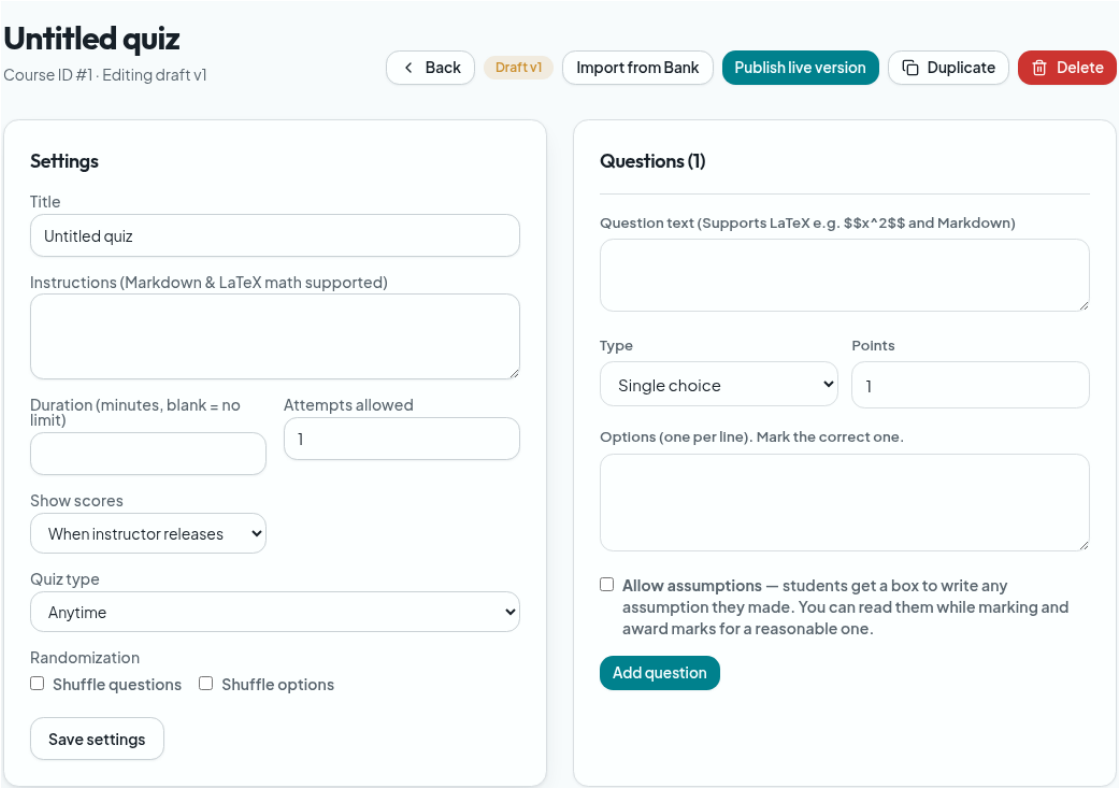


Figure 1. The saved editor view shows settings, question inputs, and the publish control. It illustrates the interface under discussion; the run records, not this image, supply the completion result. *Detail cropped from instructor-quiz-editor.png; original retained.*

05 Inspection findings

The recorded inspection uses a 0–4 severity scale: 0 indicates no issue assigned, 1 cosmetic, 2 minor, 3 major, and 4 catastrophic. These judgements do not measure the percentage of people affected. Issue priorities such as P0 are a different scale.

Table 4. Documented ratings from the inspection notes; not a new independent evaluation.

Heuristic	Lab 7	Lab 9
Visibility of system status	2	0
Match with the real world	0	0
User control and freedom	1	0
Consistency and standards	0	0
Error prevention	3	0
Recognition rather than recall	0	0
Flexibility and efficiency	1	1
Aesthetic and minimalist design	0	0
Error diagnosis and recovery	3	0
Help and documentation	1	0

5 ratings decrease. The baseline contains 2 major, 1 minor, 3 cosmetic, and 4 zero-score entries. Lab 9 contains 1 cosmetic and 9 zero-score entries. The remaining H7 note concerns editor keyboard shortcuts.

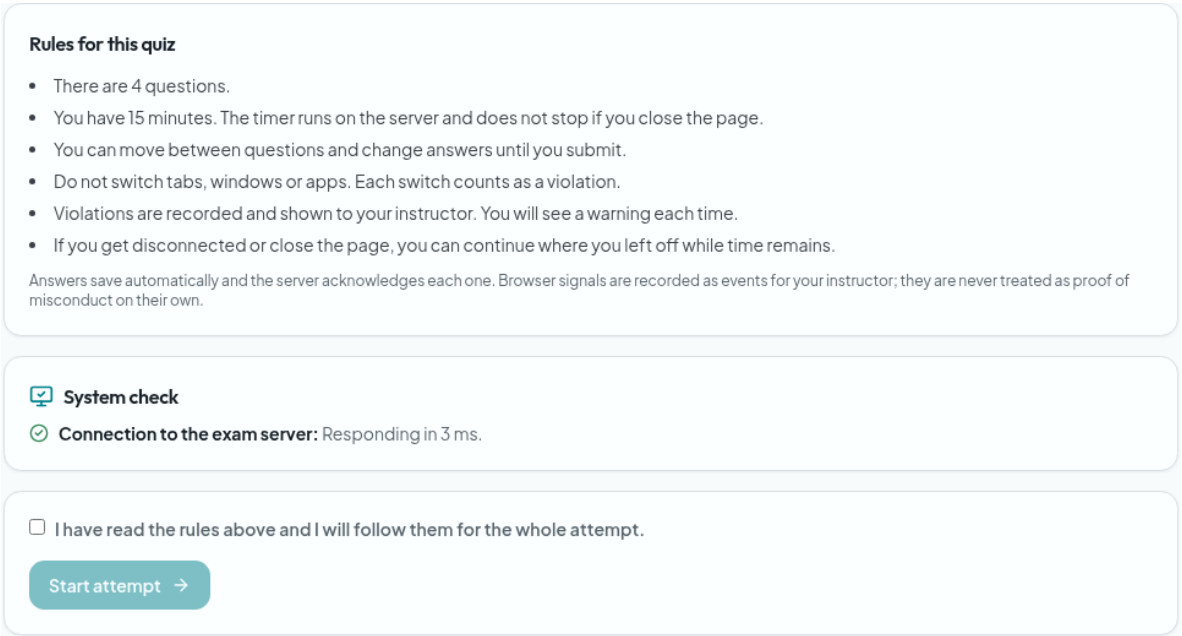


Figure 2. Preflight rules and acknowledgement are visible in the saved student view. Their presence supports the inspection's discussion of disclosure, not a claim that participants understood the rules. *Detail cropped from student-preflight.png; original retained.*

06 Accessibility: measured scope

The retained custom audit contains **55 checks: 49 passed and 6 failed**. Each failed check concerns a missing skip-to-content link. The script reports zero “hard failures” because it excludes skip-link checks from that category. This label does not mean that accessibility barriers are absent.

Table 5. Recorded checker output. A pass is specific to the test implemented, not a WCAG conformance decision.

Screen recorded by harness	Checks	Pass	Fail
Sign-in	9	8	1
Student dashboard	10	9	1
Student course	9	8	1
Student preflight	9	8	1
Instructor dashboard	9	8	1
Instructor editor / course	9	8	1
Total	55	49	6

What these checks do not establish

Colour contrast. The parser handles RGB and hexadecimal colours but not OKLCH. Unsupported colour samples are skipped. The retained pass values therefore do not demonstrate that every relevant text/background pair was measured.

Visible focus and keyboard access. The focus check can pass when an element receives focus, without proving that its indicator is visible. A sample of interactive elements does not exercise complete keyboard task paths or focus order.

Status announcements. Live-region presence is informational in this script. An alert attribute alone does not establish that a screen reader announces the right message at the right time. No saved assistive-technology session establishes that result.

Outstanding checks. Add and test a skip link, review focus at attempt start, measure contrast with support for the actual CSS colour space, and conduct keyboard-only and screen-reader task checks. These are substantive verification tasks, not a claim that the remaining work is merely cosmetic.

07 Discussion and remaining work

The saved executions show that both scripted paths can complete in the recorded setup. That supports the reported repair of the earlier workflow blockers. It does not establish the absence of other regressions, the quality of every error path, or usability for an independently recruited group.

The Lab 9 notes identify two further changes: the dashboard error banner now uses role=alert (I-A), and session handling avoids clearing a token merely because an authentication request was aborted (I-F). Their implementation is inspectable in the working tree [6]; a separate accessibility session is still needed to assess actual announcements.

Next validation should retain the automated regression checks, resolve the documented keyboard/focus items, and observe real participants attempting the two tasks without coaching. Report their outcomes and timings separately from the browser harness. No questionnaire scores or participant quotations are available here.

08 Evidence and reproducibility

Table 6. Paths are relative to the repository root. Raw evidence is retained unchanged.

Ref.	Evidence	Repository path
1	Raw execution records	docs/lab7/results.json docs/lab9/results.json
2	Task harness	e2e/lab9-usertest.spec.ts
3	Accessibility evidence	docs/lab9/a11y.json e2e/lab9-a11y.spec.ts
4	Calculation and plan	docs/lab9/compare.py docs/lab9/evaluation-plan.md
5	Inspection and prior work	docs/lab9/inspection-findings.md docs/lab8/improvements.md docs/lab5/user-tasks.md
6	Implementation changes	apps/web/src/auth.tsx apps/web/src/pages/DashboardPage.tsx
7	Original screen captures	docs/lab9/screenshots/